# ITA0612-Machine Learning

## LAB PRACTICE PROGRAMS

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## Outputs:

## Question 1:

```
seats = int(input("Number of seats: "))
category = input("Category (silver/gold): ").lower()

if category == "gold":
    rate = 250
else:
    rate = 150

total = seats * rate

if seats >= 5:
    total *= 0.9   # 10% discount

print("Total Ticket Cost = ₹", total)
```

```
Number of seats: 4
Category (silver/gold): gold
Total Ticket Cost = ₹ 1000
```

## Question 2:

```
stock = int(input("Current stock: "))
minimum = int(input("Minimum stock level: "))

if stock < minimum:
    print("Low stock alert! Reorder required.")
else:
    print("Stock level is sufficient.")
```

```
Current stock: 50
Minimum stock level: 46
Stock level is sufficient.
```

## Question 3:

```
age = int(input("Age: "))
weight = float(input("Weight (kg): "))
hb = float(input("Hemoglobin (g/dL): "))

if 18 <= age <= 60 and weight >= 50 and hb >= 12:
    print("Eligible for blood donation")
else:
    print("Not eligible")
```

```
Age: 55
Weight (kg): 78
Hemoglobin (g/dL): 89
Eligible for blood donation
```

## Question 4:

```
password = input("Enter password: ")

has_upper = any(c.isupper() for c in password)
has_lower = any(c.islower() for c in password)
has_digit = any(c.isdigit() for c in password)
has_special = any(not c.isalnum() for c in password)

if len(password) >= 8 and has_upper and has_lower and has_digit and has_special:
    print("Strong Password")
else:
    print("Weak Password")
```

```
Enter password: .asahHb231
Strong Password
```

## Question 5:

```python
age = int(input("Enter age: "))
nationality = input("Enter nationality: ").lower()

if age >= 18 and nationality == "indian":
    print("Eligible to vote")
else:
    print("Not eligible")
```

```
Enter age: 19
Enter nationality: Indian
Eligible to vote
```

## Question 6:

```python
temps = []

for i in range(7):
    t = float(input(f"Temperature Day {i+1}: "))
    temps.append(t)

print("Maximum:", max(temps))
print("Minimum:", min(temps))
print("Average:", sum(temps)/7)
```

```
Temperature Day 1: 90
Temperature Day 2: 84
Temperature Day 3: 83
Temperature Day 4: 87
Temperature Day 5: 76
Temperature Day 6: 85
Temperature Day 7: 85
Maximum: 90.0
Minimum: 76.0
Average: 84.28571428571429
```

## Question 7:

```
attended = int(input("Classes attended: "))
total = int(input("Total classes: "))

percentage = (attended / total) * 100

print("Attendance %:", round(percentage, 2))

if percentage >= 75:
    print("Eligible for exam")
else:
    print("Not eligible for exam")
```

```
Classes attended: 17
Total classes: 25
Attendance %: 68.0
Not eligible for exam
```

## Question 8:

```
vehicle = input("Vehicle type (bike/car): ").lower()
hours = int(input("Parking hours: "))

if vehicle == "bike":
    fee = hours * 10
else:
    fee = hours * 20

print("Parking Fee = ₹", fee)
```

```
Vehicle type (bike/car): car
Parking hours: 3
Parking Fee = ₹ 60
```

**Question 9:**

```
recharge = float(input("Recharge amount: "))

cashback = 0.10 * recharge
final = recharge - cashback

print("Cashback:", cashback)
print("Final amount:", final)
```

```
Recharge amount: 300
Cashback: 30.0
Final amount: 270.0
```

**Question 10:**

```
distance = float(input("Distance (km): "))
mileage = float(input("Mileage (km/l): "))
price = float(input("Fuel price per litre: "))

litres = distance / mileage
cost = litres * price

print("Fuel needed:", round(litres, 2), "L")
print("Fuel cost = ₹", round(cost, 2))
```

```
...
Distance (km): 250
Mileage (km/l): 30
Fuel price per litre: 84
Fuel needed: 8.33 L
Fuel cost = ₹ 700.0
```

## Question 11:

```
age = int(input("Enter age: "))
fare = 500

if age < 5:
    fare = 0
elif age >= 60:
    fare *= 0.5

print("Ticket Fare = ₹", fare)
```

```
Enter age: 19
Ticket Fare = ₹ 500
```

## Question 12:

```
days = int(input("Enter overdue days: "))

fine = days * 2
print("Library Fine = ₹", fine)
```

```
Enter overdue days: 2
Library Fine = ₹ 4
```

## Question 13:

```
weight = float(input("Enter weight (kg): "))
height = float(input("Enter height (m): "))

bmi = weight / (height ** 2)

if bmi < 18.5:
    status = "Underweight"
elif bmi < 25:
    status = "Normal"
elif bmi < 30:
    status = "Overweight"
else:
    status = "Obese"

print("BMI:", round(bmi, 2))
print("Status:", status)
```

```
Enter weight (kg): 64
Enter height (m): 3
BMI: 7.11
Status: Underweight
```

## Question 14:

```
p = float(input("Principal: "))
r = float(input("Rate (%): "))
t = float(input("Time (years): "))

si = (p * r * t) / 100
ci = p * (1 + r/100) ** t - p

print("Simple Interest:", si)
print("Compound Interest:", ci)
```

```
Principal: 2000
Rate (%): 2
Time (years): 3
Simple Interest: 120.0
Compound Interest: 122.41600000000017
```

## Question 15:

```
food = float(input("Enter food amount: "))

gst = 0.05 * food
service = 0.10 * food
total = food + gst + service

print("GST:", gst)
print("Service Charge:", service)
print("Total Bill:", total)
```

```
Enter food amount: 200
GST: 10.0
Service Charge: 20.0
Total Bill: 230.0
```

## Question 16:

```
amount = float(input("Enter purchase amount: "))

if amount > 5000:
    discount = 0.20 * amount
elif amount > 2000:
    discount = 0.10 * amount
else:
    discount = 0

subtotal = amount - discount
gst = 0.18 * subtotal
final = subtotal + gst

print("Discount:", discount)
print("GST:", gst)
print("Final Bill:", final)
```

```
Enter purchase amount: 7000
Discount: 1400.0
GST: 1008.0
Final Bill: 6608.0
```

## Question 17:

```python
correct_pin = "1234"
pin = input("Enter PIN: ")
balance = float(input("Enter account balance: "))
amount = float(input("Enter withdrawal amount: "))

if pin == correct_pin:
    if amount <= balance:
        balance -= amount
        print("Withdrawal successful")
        print("Remaining balance:", balance)
    else:
        print("Insufficient balance")
else:
    print("Invalid PIN")
```

```
Enter PIN: 1234
Enter account balance: 10000
Enter withdrawal amount: 2000
Withdrawal successful
Remaining balance: 8000.0
```

## Question 18:

```python
# Electricity Bill Generator

print("=== Electricity Bill Generator ===")

units = float(input("Enter Electricity Units Consumed: "))

bill = 0

if units <= 100:
    bill = units * 1.5
elif units <= 200:
    bill = (100 * 1.5) + ((units - 100) * 2.5)
elif units <= 300:
    bill = (100 * 1.5) + (100 * 2.5) + ((units - 200) * 4)
else:
    bill = (100 * 1.5) + (100 * 2.5) + (100 * 4) + ((units - 300) * 6)

print("\n----- Electricity Bill -----")
print("Units Consumed:", units)
print("Total Bill: ₹", bill)
```

```
=== Electricity Bill Generator ===
Enter Electricity Units Consumed: 100

----- Electricity Bill -----
Units Consumed: 100.0
Total Bill: ₹ 150.0
```

Start

## Question 19:

```python
# Allowances
hra = 0.20 * basic_salary      # 20%
da = 0.10 * basic_salary       # 10%

gross_salary = basic_salary + hra + da

# Deductions
pf = 0.12 * basic_salary       # 12%
tax = 0.05 * gross_salary      # 5%

net_salary = gross_salary - (pf + tax)

# Display
print("\n----- Salary Details -----")
print("Basic Salary :", basic_salary)
print("HRA :", hra)
print("DA :", da)
print("Gross Salary :", gross_salary)
print("PF Deduction :", pf)
print("Tax :", tax)
print("Net Salary :", net_salary)
```

```
=== Employee Salary Calculator ===
Enter Basic Salary: 30000

----- Salary Details -----
Basic Salary : 30000.0
HRA : 6000.0
DA : 3000.0
Gross Salary : 39000.0
PF Deduction : 3600.0
Tax : 1950.0
Net Salary : 33450.0
```

## Question 20:

```
total = m1 + m2 + m3 + m4 + m5
average = total / 5

if average >= 90:
    grade = "A"
elif average >= 80:
    grade = "B"
elif average >= 70:
    grade = "C"
elif average >= 60:
    grade = "D"
elif average >= 50:
    grade = "E"
else:
    grade = "F"

print("\n----- Result -----")
print("Total Marks:", total)
print("Average Marks:", average)
print("Grade:", grade)
```

```
Student Grade Calculator
Enter marks for Subject 1: 82
Enter marks for Subject 2: 92
Enter marks for Subject 3: 82
Enter marks for Subject 4: 83
Enter marks for Subject 5: 85

----- Result -----
Total Marks: 424.0
Average Marks: 84.8
Grade: B
```